

ClimateScope GlobalWeather, Climate, and Environmental Analysis

A comprehensive Power BI dashboard for analyzing global weather patterns, climate trends, and environmental factors including air quality metrics.



Project Overview

This interactive dashboard transforms complex climate data into actionable insights. It analyzes temperature variations, precipitation patterns, wind dynamics, humidity levels, and air quality metrics across countries and time periods.

Designed for stakeholders, researchers, and decision-makers who need to understand environmental patterns and make data-driven decisions about climate-related initiatives.



Three-Page Structure



Over view

Global temperature distribution, weather patterns, and rainfall by country with summary KPIs



Temperature & Climate

Detailed temperature analysis by weather condition, humidity distribution, and wind patterns



Air Quality & Environment

Pollution metrics, visibility analysis, and environmental impact assessment

Project Objectives

01

Global Weather Summary

Provide a high-level view of worldwide weather conditions and climate patterns

03

Environmental Assessment

Analyze air quality factors including PM2.5, PM10, NO2, CO, and UV Index

02

Climate Analysis

Enable detailed examination of temperature, humidity, wind speed, and weather conditions

04

Data-Driven Decisions

Support informed decision-making through interactive, visual analytics

Dataset Structure

Weather Metrics

- Temperature (Celsius)
- Wind Speed (Kph)
- Precipitation (mm)
- Humidity (%)
- Weather Condition
- FeelsLike Temperature

Environmental Factors

- PM2.5, PM10 levels
- NO2 and CO concentration
- UV Index
- Visibility (km)



Geographic Context

- Country/Region
- Date and Time
- Location coordinates

Page 1: Overview

KPI Cards

Average temperature, maximum wind speed, total rainfall, and average humidity displayed as summary metrics

Global Temperature Map

Interactive world map showing temperature distribution by country and region

Weather Distribution

Chart showing frequency of different weather conditions across the dataset

Rainfall by Country

Bar chart comparing precipitation levels across different nations

Scatter Plot

Humidity versus wind speed correlation analysis

Temperature Trend

Line chart showing temperature changes over time

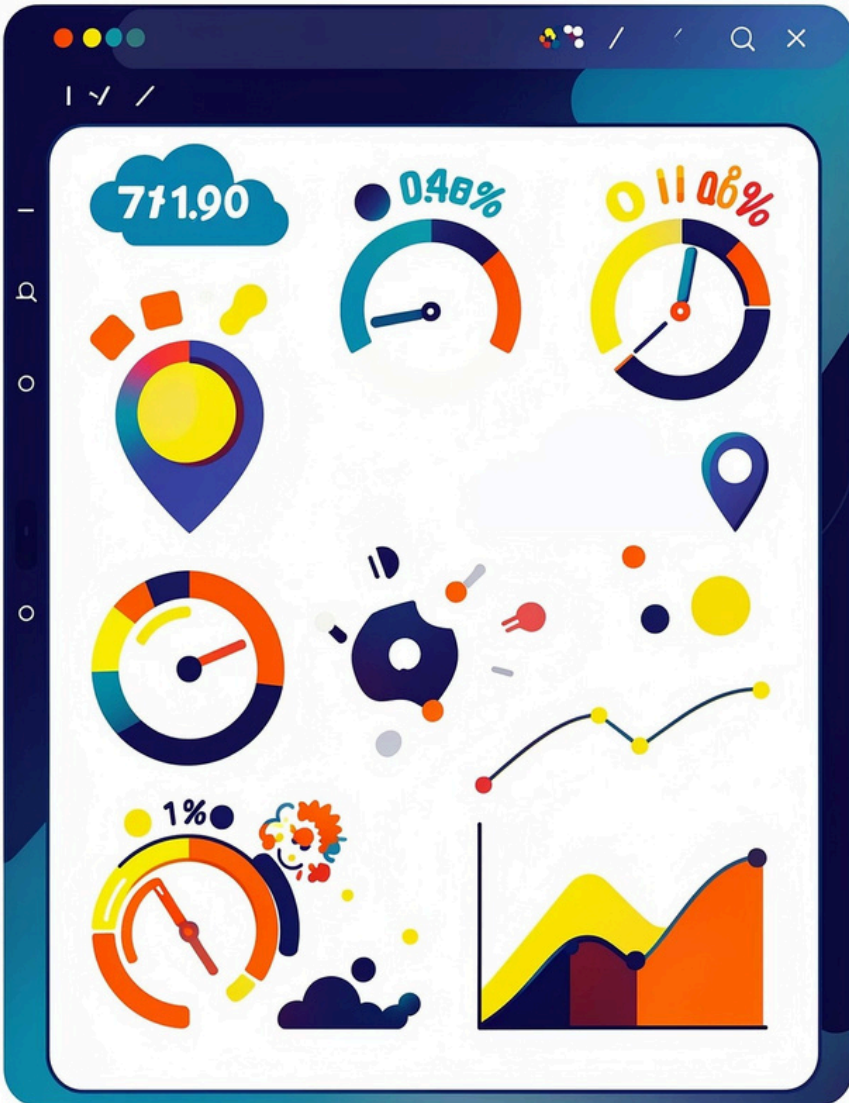
Pages 2-3: Detailed Analysis

Temperature & Climate

- Temperature by weather condition
- Temperature distribution histogram
- Humidity distribution analysis
- Wind speed by direction
- Feels like vs actual temperature comparison

AirQuality & Environment

- Pollution metrics dashboard
- Pollution by country comparison
- Visibility vs pollution correlation
- Rainfall impact on pollution
- UV Index comparison across regions



Design & Modeling Principles



Visual Design

Clean, minimal UI with consistent blue, white, and grey color palette. Cards with shadows provide visual separation and hierarchy.



Data Modeling

Key aggregations include averages for temperature and humidity, sum for rainfall, and count for distributions. Bins enable histogram creation.



Layout Structure

Logical flow from top (summary KPIs) to middle (main visuals) to bottom (detailed analysis). Avoids clutter through strategic visual selection.



User Interaction

Slicers for weather type, country, and date enable interactive filtering. Drill-down capabilities provide deeper insights where applicable.

Key Insights



Temperature Variance

Significant temperature variations observed across different weather conditions, with clear patterns emerging by region



Humidity Correlation

Higher humidity levels strongly correlate with rainfall patterns, providing predictive value for precipitation



Visibility Impact

Pollution levels directly impact visibility metrics, with PM2.5 showing strongest correlation to reduced visibility



Cleansing Effect

Rainfall events demonstrate measurable reduction in pollution levels, particularly for PM10 and PM2.5 particles

Conclusion

This ClimateScope dashboard transforms complex environmental data into clear, actionable insights. By combining weather patterns, climate trends, and air quality metrics in an interactive format, it enables stakeholders to understand global climate patterns and make informed decisions.

The three-page structure provides both high-level summaries and detailed analysis, while consistent design principles ensure clarity and usability. Interactive elements empower users to explore data relevant to their specific needs.



3

Dashboard Pages

15+

Interactive charts and maps

10+

Complete metrics set